

WEEK 1

Task1

Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that

1. Accept a Linux command as input.
2. Create a child process using fork().
3. Execute the command in the child process using an appropriate exec() system call.
4. Allow the parent process to wait for the child using wait ().
5. Display the Process ID (PID) of both parent and child processes.

Source Code:

```
#include <stdio.h>

#include <stdlib.h>

#include <unistd.h>

#include <sys/types.h>

#include <sys/wait.h>

#include <string.h>

int main(){

    char command[100];

    pid_t pid;

    printf("Enter a Linux command: ");

    scanf("%s", command);

    pid = fork();

    if(pid < 0) {

        printf("Fork failed!\n");

        return 1;

    }

    else if(pid == 0) {

        printf("\nChild Process\n");

        printf("Child PID : %d\n", getpid());
```

```
    execlp(command, command, NULL);  
    printf("Command execution failed.\n");  
    exit(1);  
} else {  
    wait(NULL);  
    printf("\nParent Process\n");  
    printf("Parent PID : %d\n", getpid());  
    printf("Child process completed.\n");  
}  
return 0;  
}
```

Output:

```
Enter a Linux command: ls  
  
Child Process  
Child PID : 437  
OSSP  code.c  files.c  gm.c    hello.c  os      program.c  programming.c  source_code  
code  files   gm       hello   operating os.c     programming  programming.c.save  source_code.c  
  
Parent Process  
Parent PID : 436  
Child process completed.
```

Task2

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

Report

1. Uname: Displays operating system and kernel information.

```
root@DESKTOP-VMN932K:~# uname  
Linux
```

2. lscpu: Displays CPU information.

```
root@DESKTOP-VMN932K:~# lscpu  
Architecture:          x86_64  
CPU op-mode(s):        32-bit, 64-bit  
Address sizes:          39 bits physical, 48 bits virtual  
Byte Order:            Little Endian  
CPU(s):                4  
On-line CPU(s) list:    0-3  
Vendor ID:              GenuineIntel  
Model name:             Intel(R) Core(TM) i3-8145U CPU @ 2.10GHz  
CPU family:             6  
Model:                  142  
Thread(s) per core:     2  
Core(s) per socket:     2  
Socket(s):              1  
Stepping:               12  
BogoMIPS:               4608.01
```

3. lsblk: Displays storage devices.

```
root@DESKTOP-VMN932K:~# lsblk  
NAME  
MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS  
sda  8:0    0 356.9M  1 disk  
sdb  8:16   0 159.4M  1 disk  
sdc  8:32   0    2G    0 disk [SWAP]  
sdd  8:48   0    1T    0 disk /mnt/wslg/distro  
/
```

4. ps: Displays currently running processes.

```
root@DESKTOP-VMN932K:~# ps  
PID TTY          TIME CMD  
476 pts/0        00:00:00 bash  
505 pts/0        00:00:00 ps
```

5. top: Displays real-time system performance.

```
root@DESKTOP-VMN932K:~# top
top - 13:56:44 up 22 min,  1 user,  load average: 0.06, 0.03, 0.00
Tasks:  22 total,   1 running, 21 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,   0.0 wa,   0.0 hi,   0.0 si,   0.0 st
MiB Mem :  7880.6 total,  7463.6 free,   438.7 used,   123.2 buff/cache
MiB Swap:  2048.0 total,  2048.0 free,    0.0 used.  7441.9 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	21680	13140	9744	S	0.0	0.2	0:00.74	systemd
2	root	20	0	3180	2200	2072	S	0.0	0.0	0:00.01	init-systemd(Ub
9	root	20	0	3196	2128	2008	S	0.0	0.0	0:00.00	init
64	root	19	-1	34052	12556	11488	S	0.0	0.2	0:00.25	systemd-journal
112	root	20	0	25548	6924	5168	S	0.0	0.1	0:00.26	systemd-udev
123	systemd+	20	0	21344	13252	11124	S	0.0	0.2	0:00.12	systemd-resolve
124	systemd+	20	0	91036	7992	7024	S	0.0	0.1	0:00.10	systemd-timesyn
168	root	20	0	4244	2688	2432	S	0.0	0.0	0:00.00	cron
169	message+	20	0	9540	5432	4736	S	0.0	0.1	0:00.09	dbus-daemon
176	root	20	0	17980	8888	7848	S	0.0	0.1	0:00.08	systemd-logind
178	root	20	0	1756104	13024	10888	S	0.0	0.2	0:00.14	wsl-pro-service
183	root	20	0	333516	5068	1564	S	0.0	0.1	0:00.00	systemd-logind